

Strategic Games: Introduction

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Decision Theory

- Decision theory can be divided in two two large class of economic settings:
 - individual (non-strategic) decision theory
 - payoff of the agent depends only on his/her choices (e.g. *consumer theory*, *producer theory*, *partial/general equilibrium analysis*).
 - multi-person (strategic) decision theory
 - payoff of agents depends not only on their own actions, but also on the actions of other agents (e.g. *taxpayer-IRS*, *auctions*, *public good provision*, *oligopolistic competition*...)

Game Theory: a methodology used to study the interaction of agents in strategic situations.

Cooperative Game Theory (M (Ch.18-Appendix))

- *Cooperative Game Theory*: a theory which assumes that agents can attain certain outcomes through binding agreements. The goal is to provide characteristics of agreements that are likely to emerge.

Example: market interaction as cooperative game

Consider an economy with

- I consumers with increasing and convex utility functions $u_i : \mathbb{R}_+^I \rightarrow \mathbb{R}$ and endowments $\omega_i \geq 0$.
- $Y \subset \mathbb{R}^I$ convex, constant-return technology.

Let $s \subset I$ denote a possible coalition structure (a partition of consumers). We can define a cooperative game (in characteristic form) as $V(s) \subset \mathbb{R}^s$ such that

$$V(s) = \{(u_i(x_i))_{i \in s} : \sum_{i \in s} x_i = \sum_{i \in s} \omega_i + y, y \in Y\}$$

Cooperative Game Theory (M (Ch.18-Appendix))

- The goal is to determine which outcomes $O(S) \subset V(S)$ where $S \subset \mathcal{P}(I)$ ($\mathcal{P}(I)$ is the set of all possible partitions) are likely to emerge.

Example: market interaction and the core

Suppose that $I = 2$.

- How can we represent $V(S)$ and $O(S)$ graphically?
- Can we narrow our prediction regarding the outcomes $O^c(s) \subset O(S)$ that are likely to emerge as a cooperative solution?

Non-cooperative Game Theory

- *Non-cooperative Game Theory*: agents make decisions independently and any agreements must be self-enforcing.
- Non-cooperative games can take many forms.
 - Timing:
 - static (one-shot) games;
 - dynamic (multi-stage) games;
 - Information structure:
 - complete (incomplete) information: each player's payoff function is common knowledge (privately known);
 - perfect (imperfect) information: each player knows (does not know) the full history of the game when choosing his/her move;

Static Games of Complete and Perfect Information (SGCPI)

- A strategic (normal) form game $\Gamma = \{I, S, u\}$ consists of three components:
 - Finite set of players $I \equiv \{1, 2, \dots, I\}$
 - Set of strategies $S_i = (s_i^1, s_i^2, \dots, s_i^n)$ for each player $i \in I$.
 - $s \equiv (s_1, s_2, \dots, s_I)$ denotes a strategy profile specifying one strategy for each player.
 - Then $S \equiv \times_{i=1}^I S_i$ describes all possible strategy profiles.
 - Payoffs for each player i , $u_i(s) : S \rightarrow \mathbb{R}$.
- In SGCPI, each player $i \in I$ simultaneously selects a strategy s_i from his set S_i giving rise to a payoff profile $(u_1(s), u_2(s), \dots, u_I(s))$.

Example: Prisoner's dilemma

- 2 players;
- 2 strategies: *C*-confess, *NC*- do not confess;

		Player 2	
		<i>C</i>	<i>NC</i>
Player 1	<i>C</i>	-9, -9	0, -10
	<i>NC</i>	-10, 0	-1, -1

Figure: Prisoner's dilemma 1

Full Equivalence

Two normal form games $\Gamma = \{I, S, u\}$ and $\tilde{\Gamma} = \{I, S, \tilde{u}\}$ are fully equivalent if for all i , there exists $A_i > 0$ and B_i such that $\tilde{u}_i(s) = A_i u_i(s) + B_i$.

- The following game is equivalent to the Prisoner's dilemma game.

		Player 2	
		C	NC
Player 1	C	0, 0	9, -1
	NC	-1, 9	8, 8

Figure: Prisoner's dilemma 2

Mixed Strategies

A mixed strategy for player i is a function $\sigma_i : S_i \rightarrow [0, 1]$, which assigns probability $\sigma_i(s_i) \geq 0$ to each pure strategy $s_i \in S_i$, satisfying $\sum_{s_i \in S_i} \sigma_i(s_i) = 1$. Player i payoff from profile σ is

$$u_i(\sigma) = \sum_{s \in S} \left(\prod_{j=1}^I \sigma_j(s_j) \right) u_i(s)$$

Σ_i denotes the space of i 's mixed strategies.

- Example: In the Prisoner's dilemma 1, if both players adopt $\sigma_i(C) = \sigma_i(NC) = \frac{1}{2}$, then the expected payoff for each player i is:

$$u_i(\sigma) = -9 \times \frac{1}{4} - 10 \times \frac{1}{4} - 1 \times \frac{1}{4} = -5$$

Dominant Strategies

- Notation: $s_{-i} = (s_1, \dots, s_i, s_{i+1}, \dots, s_I)$

Dominant Strategy

A strategy $s_i \in S_i$ is strictly dominant for i , if for all $s'_i \in S_i \setminus \{s_i\}$ and all $s_{-i} \in S_{-i}$

$$u_i(s_i, s_{-i}) > u_i(s'_i, s_{-i})$$

- Question:
 - Why don't we consider mixed strategies?
 - Why is it sufficient to for a dominant strategy to generate a higher payoff than any other strategy for only *pure* strategies of the other players?
 - Does Prisoner's dilemma have a dominant strategy for some player?

Dominated Strategies

Dominated Strategy

A strategy σ_i is strictly dominated for player i if there exists $\sigma'_i \in \Sigma_i$ such that for all $s_{-i} \in S_{-i}$

$$u_i(\sigma'_i, s_{-i}) > u_i(\sigma_i, s_{-i})$$

- Why should we consider mixed strategies in this case?
- Can an undominated mixed strategy contain a dominated pure strategy?
- Can a mixed strategy that only contains undominated pure strategies be dominated?

Example

- Are there dominated pure strategies in the following game?

		Player 2	
		L	R
Player 1	U	1, 3	-0.5, 0
	M	-0.5, 0	1, 3
	D	0, 1	0, 1

Figure: Game 3

Example: Iterated Strict Dominance

- *Iterated strict dominance*: a process of iteratively eliminating dominated strategies.
- Consider the following game:

		Player 2		
		L	C	R
Player 1	U	0, 0	3.1, 2	4.6, 2.3
	M	2, 3.1	4.1, 4.1	5.1, 3.8
	D	2.3, 4.6	3.8, 5.1	4.6, 4.6

Figure: Game 4

Definition: Iterative Strict Dominance (ISD)

Set $S_i^0 \equiv S_i$ and $\Sigma_i^0 \equiv \Sigma_i$. Define S_i^n recursively as follow:

$$S_i^n = \{s_i \in S_i^{n-1} \mid \text{there is no } \sigma_i \text{ s.t. } u_i(\sigma_i, s_{-i}) > u_i(s_i, s_{-i}) \text{ for all } s_{-i} \in S_{-i}^{n-1}\}$$

and define

$$\Sigma_i^n = \{\sigma_i \in \Sigma_i \mid \sigma_i(s_i) > 0 \text{ only if } s_i \in S_i^n\}$$

Set $S_i^\infty = \bigcap_{n=0}^\infty S_i^n$ and $S^\infty \equiv \times_{i=1}^I S_i^\infty$. Similarly, $\Sigma^\infty \equiv \times_{i=1}^I \Sigma_i^\infty$.

$$\Sigma_i^\infty = \{\sigma \in \Sigma_i \mid \text{there is no } \sigma'_i \text{ s.t. } u_i(\sigma'_i, s_{-i}) > u_i(\sigma_i, s_{-i}) \text{ for all } s_{-i} \in S_{-i}^\infty\}$$

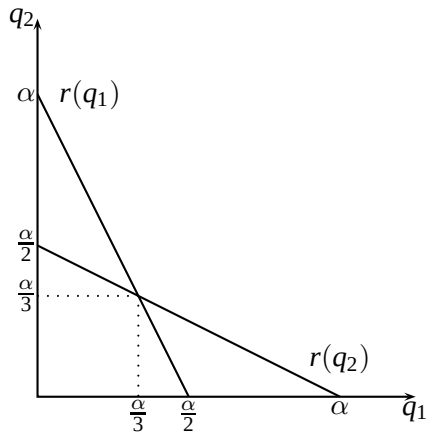
ISD as a solution concept

- A game is solvable by iterated strict dominance if, for each player i , S_i^∞ is a singleton (i.e., a one element set).
- Example: Cournot Game
 - Players: firm 1 and firm 2;
 - Simultaneously choose their output levels (q_1, q_2) where $q_i \geq 0$ and the output is perfectly divisible (for simplicity);
 - The inverse market demand is $P(Q) = \alpha - Q$ where $Q = q_1 + q_2$;
 - Cost of production is 0.
 - Therefore, the profit for firm i is:

$$\max_{q_i} \pi_i = (\alpha - q_i - q_j)q_i$$

- For a given q_j , firm i 's optimal output level is $q_i = r_i(q_j) = \frac{\alpha - q_j}{2}$.

Reaction Functions



ISD-does the order of elimination matter?

Example (Dufwenberg and Stegeman, Econometrica 2002)

Consider a one player game with strategy set $G_1 = (0, 1)$ and a payoff function $u_1(x) = x$ for $x \in G_1$.

- Are there dominated strategies?
- Does the order of elimination of dominated strategies matter?
- If so, what is the reason behind it?

ISD- does order of elimination matter?

Example (Dufwenberg and Stegeman, Econometrica 2002)

Example: Consider a two-player game with strategy sets $S_1 = S_2 = [0, 1]$ and payoff functions $u_i : S_i \times S_j$ with $i, j = 1, 2$ and $i \neq j$

$$u_i(s_i, s_j) = \begin{cases} s_i & \text{if } s_i < 1 \\ 0 & \text{if } s_i = 1 \text{ and } s_j < 1 \\ 1 & \text{if } s_i = 1 \text{ and } s_j = 1 \end{cases}$$

- Are there dominated strategies?
- Does the order of elimination of dominated strategies matter?
- If so, what is the reason behind it?

When is ISD immune to order concerns?

Definition

We say that $\Gamma = \{I, S, u\}$ is compact and continuous if S_i is a compact set and u_i is continuous $\forall i \in I$.

Theorem 1

Let Γ be a compact and continuous game. Then any maximal reduction Σ^∞ is unique.

Proof of Theorem 1

Let Γ be a compact and continuous game and suppose that Σ^∞ is not unique meaning that there exist $\Sigma^{1,\infty}$ and $\Sigma^{2,\infty}$ such that $\Sigma^{2,\infty} \not\subseteq \Sigma^{1,\infty}$. This implies that $\exists \sigma'_i \in \Sigma_i^{2,\infty}$ such that $\sigma'_i \notin \Sigma_i^{1,\infty}$ for some i .

Since $\Sigma_i^{1,\infty} = \bigcap_{n=0}^{\infty} \Sigma_i^n$, $\exists n$ and σ_i s.t. $\sigma'_i \in \Sigma_i^n$ and $u_i(\sigma_i, s_{-i}) > u_i(\sigma'_i, s_{-i})$ for all $s_{-i} \in S_{-i}^{n-1}$. Note that Σ^n is a compact set since $\Sigma^n \subseteq \Gamma$ and Σ^n is closed. Then, since Σ^n is compact and $u_i(\cdot)$ continuous, it follows that $\exists \sigma_i^*$ such that

- 1) $u(\sigma_i^*, s_{-i}) > u_i(\sigma'_i, s_{-i}), \forall s_{-i} \in S_{-i}^{n-1}$
- 2) $u(\sigma_i^*, s_{-i}) \geq u(\sigma_i, s_{-i}), \forall \sigma_i$ and some $s_{-i} \in S_{-i}^{n-1}$.

If $\sigma_i^* \in \Sigma^{2,\infty}$, we are done since 1) implies that $\Sigma^{2,\infty}$ cannot be a maximal reduction of S and one more iteration eliminates σ'_i .

If $\sigma_i^* \notin \Sigma^{2,\infty}$, then $\exists \tilde{\sigma}_i \in \Sigma^{2,\infty}$ such that $u_i(\tilde{\sigma}_i, s_{-i}) > u_i(\sigma_i^*, s_{-i}) > u_i(\sigma'_i, s_{-i}) \forall s_{-i} \in S_{-i}^\infty$ since $S_{-i}^\infty \subseteq S_{-i}^{n-1}$ for any n . This again implies that $\Sigma^{2,\infty}$ cannot be a maximal reduction of Σ .

Therefore, it follows that $\Sigma^{1,\infty} \sim \Sigma^{2,\infty}$.

Weaknesses of ISD

- Might not be immune to the order of elimination (although it holds for a large class of games).
- The process of deletion does not stop after a finite number of steps. Each round of elimination requires one deeper level of mutual knowledge that players are rational.

Mutual and Common Knowledge

A fact F is a mutual knowledge of order t if 1) all players know F ; 2) all players know that all players know F ; up to t iterations. F is a common knowledge if F is a mutual knowledge of order ∞ .

- conclusions based on a large number of iterations are usually less robust to small changes in information that players have regarding each other's rationality.
- The maximal reduction set can still be quite large.

Best response

Definition

A mixed strategy σ_i is a best response for player i against some σ_{-i} if

$$u_i(\sigma_i, \sigma_{-i}) \geq u_i(\sigma'_i, \sigma_{-i}) \quad \forall \sigma'_i \in \Sigma_i \setminus \sigma_i$$

A strategy σ_i is never a best response if there is no $\sigma_{-i} \in \Sigma_{-i}$ for which it is a best response.

- Another approach we can take is simply removing strategies that are never best response.

Rationalizability

Definition

Set $\tilde{\Sigma}_i^0 \equiv \Sigma_i$ and for each i recursively define

$\tilde{\Sigma}_i^n = \{\sigma_i \in \tilde{\Sigma}_i^{n-1} \mid \exists \sigma_{-i} \in \times_{j \neq i} \text{convex hull}(\tilde{\Sigma}_j^{n-1}) \text{ s. t. } u_i(\sigma_i, \sigma_{-i}) \geq u_i(\sigma'_i, \sigma_{-i}) \text{ for all } \sigma'_i \in \tilde{\Sigma}_i^{n-1}\}$

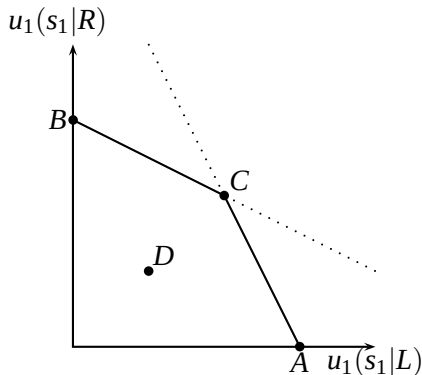
The *rationalizable strategies* for player i are $R_i = \bigcap_{n=0}^{\infty} \tilde{\Sigma}_i^n$.

- In other words, a strategy is rationalizable for player i if it survives iterative elimination of strategies that are never a best response.
- The set of rationalizable strategies is nonempty and contains at least one pure strategy for each player.
- Can a strictly dominated strategy be a best response?
- Are mixed strategies containing pure rationalizable strategies rationalizable?

Example

- Two players: 1 and 2.
- $s_1 = \{A, B, C, D\}$
- $s_2 = \{R, L\}$

s_1	$u_1(s_1 L)$	$u_1(s_1 R)$
A	3	0
B	0	3
C	2	2
D	1	1



Rationalizability and Strict Dominance

Theorem 2

Rationalizability and iterated strict dominance coincide in two-player games.

Sketch of Proof

The proof follows an inductive argument. By definition $\Sigma^0 \sim \tilde{\Sigma}^0$. Assume that $\Sigma^n \sim \tilde{\Sigma}^n$. We have to prove that it is true for $\Sigma^{n+1} \sim \tilde{\Sigma}^{n+1}$ (i.e. for all $\sigma_i \in \Sigma^{n+1}, \sigma_i \in \tilde{\Sigma}^{n+1}$). We focus on $\Sigma^{n+1} \subseteq \tilde{\Sigma}^{n+1}$ (the other direction is easy). Consider the convex set of vectors $\vec{u}_i(\sigma_i) = \{u_i(\sigma_i, s_j)\}_{s_j \in S_j^n}$ for all $\sigma_i \in \Sigma_i^n$. Let $\sigma_i \in \Sigma_i^{n+1}$. We want to show that $\sigma_i \in \tilde{\Sigma}^{n+1}$. Since σ_i is undominated in this set, it lies on the efficient frontier. By the separating hyperplane theorem, $\exists \sigma_j = \{\sigma_j(s_j)\}_{s_j \in S_j^n}$ s.t.

$$\sigma_j \cdot (\vec{u}_i(\sigma_i) - \vec{u}_i(\sigma'_i)) \geq 0$$

or $u_i(\sigma_i, \sigma_j) \geq u_i(\sigma'_i, \sigma_j) \forall \sigma_i \in \Sigma_i^n = \tilde{\Sigma}_i^n$.

References

- Mas-Colell- Chapter 18-Appendix
- Fudenberg and Tirole- Chapter 2
- Dufwenber M., and Stegeman M., “Existence and Uniqueness of Maximal Reduction under Iterated Strict Dominance”, Econometrica, 2002, Vol. 70, No. 5